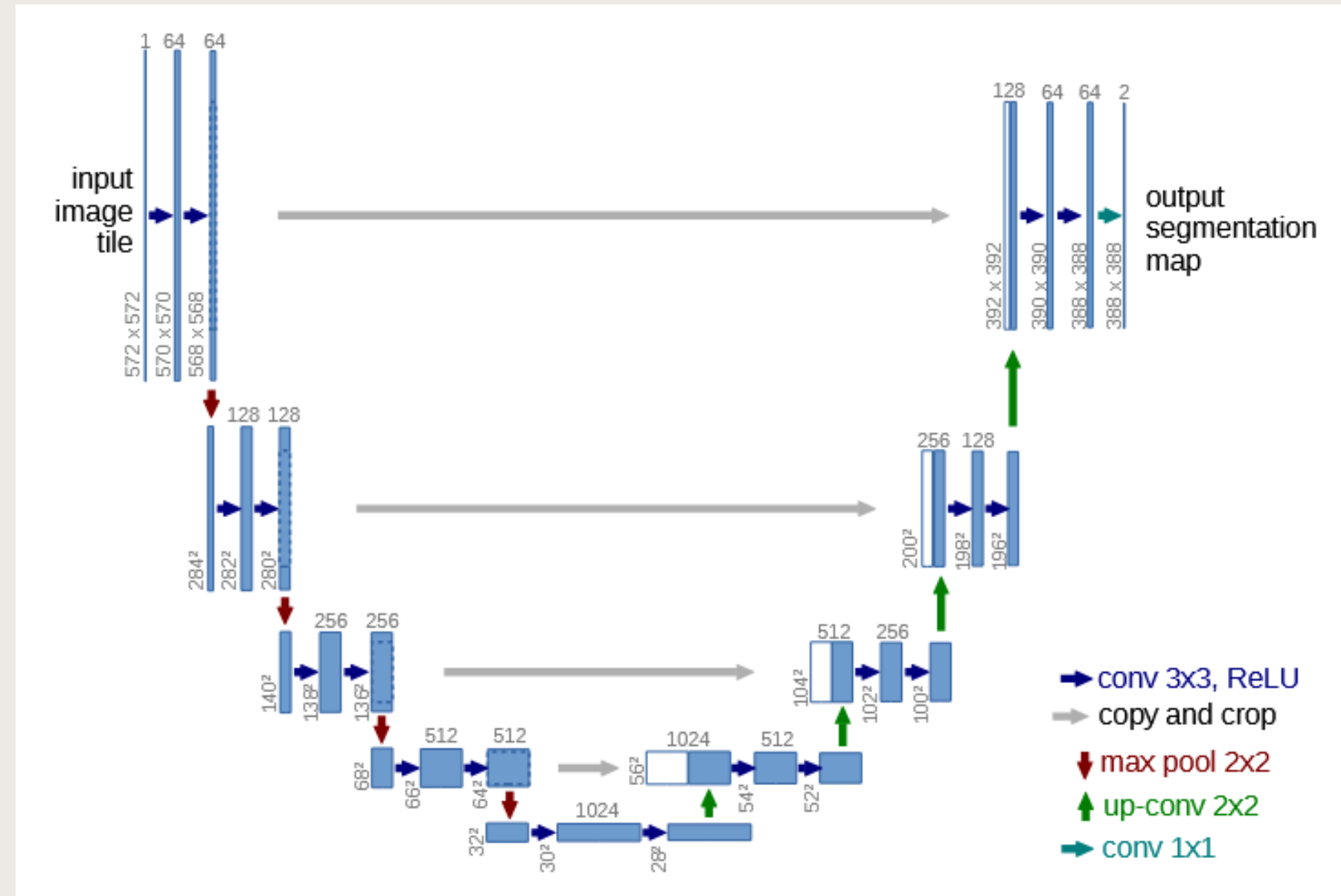


U-Net architecture



1. U-Net++:

This is an extension of the classic U-Net, featuring a dense mesh of connections between the encoder and the decoder. Using nested and stepped paths enables features from different resolution levels to be used more effectively, improving the precision of cell contours.

2. Attention U-Net:

Adds an attention gate mechanism to the skip-connection paths so the network can automatically focus on the most relevant areas (in this case, cells) while suppressing background noise. This enables more selective merging of features from the encoder.

3. RRCNN U-Net:

combines U-Net with recursive ResRCNN (Residual Recurrent CNN) blocks. Recursion allows features to be processed repeatedly, and residual linking facilitates gradient flow. This results in richer representations of cell features.

4. RRCNN + Attention:

This combines the U-Net RRCNN with an attention mechanism so that the network processes features repeatedly and residually and also selects relevant segmentation areas. The result is clearer masks and better separation of neighbouring cells.

5. Star-Convex Polygons (StarDist):

This model generates distances in specific directions for each cell point to create star-convex polygons. This accurately represents cell shapes and separates adjacent objects, even those with complex contours.

<u>Architec ture</u>	<u>Best Loss</u>	<u>Best Layers</u>	<u>MIoU</u>
<u>U-Net</u>	<u>LogCosh DiceLoss</u>	<u>1-4-16-32 -64</u>	<u>0.894</u>
<u>BiFPN</u>	<u>DiceLoss</u>	<u>1-4-16-32 -64</u>	<u>0.899</u>
<u>Attention U-Net</u>	<u>DiceLoss</u>	<u>1-4-16-32 -64</u>	<u>0.890</u>
<u>RR U-Net</u>	<u>DiceLoss</u>	<u>1-4-16-32 -64</u>	<u>0.890</u>
<u>U-Net++</u>	<u>DiceLoss</u>	<u>1-4-16-32 -64</u>	<u>0.878</u>
<u>AttRRU- Net</u>	<u>Combo- Loss</u>	<u>1-4-8-16- 32</u>	<u>0.883</u>

BiFPN

A BiFPN, or Weighted Bi-directional Feature Pyramid Network, is a type of feature pyramid network which allows easy and fast multi-scale feature fusion.

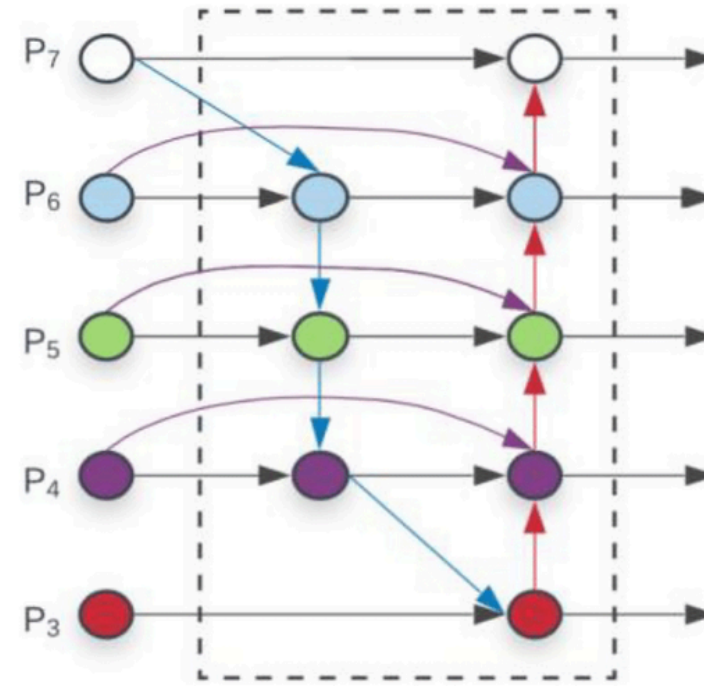


Fig. 1 Weighted Bi-directional Feature Pyramid Network

This BiFPN-enhanced U-Net replaces the standard decoder's simple upsampling and concatenation steps with learnable, attention-based fusion blocks. At each decoding level, the upsampled feature map and its corresponding encoder skip connection are aligned in the channel dimension before being merged using two softmax-normalised weights that adaptively balance their contributions.

The fused map is then processed through a series of three 3×3 convolution-batch normalization layers with dropout for regularisation, and a local skip from the first convolution layer helps preserve early fused features. By learning how much emphasis to give to high-resolution encoder details versus deeper decoder context, the network produces more precise, contextually informed feature maps at every scale.

- AdvancedAugmentationPipeline (grid-searched parameters)
- Spatial (prob = 0.0–0.5): with probability spatial_prob apply one of:
 - Rotate by a random angle in (−90°, +90°)
 - Elastic deform using a random grid size (4–8) and σ (8–16)
 - Flip horizontally or vertically
 - Color (prob = 0.0–0.25): with probability color_prob apply one of:
 - Contrast scale factor drawn from $1 \pm (0.1-0.2)$
 - Brightness shift up to $\pm(0.1-0.2) \cdot 255$
 - Noise (prob = 0.0–0.25): with probability noise_prob add Gaussian noise ($\sigma \approx 25$)

All probabilities and parameter ranges were optimized via grid search to maximize segmentation robustness.

In addition, we conducted numerous experiments to identify the optimal hyperparameters for each model. We employed the search grid and ClearML to store the results, while testing various parameters such as the number of epochs, learning rate, batch size, different cost functions, dropout and weight initialisation.

Dataset	Best Architecture	Augmentation	MIoU
DIC			
FL			
PC			

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